Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date: 30 September 2026

Team ID: _____

Project Name: Script-Controlled ACL – Restrict Record Access Based on Field Value

Maximum Marks: 4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution. The solution uses a script-controlled Access Control List (ACL) to determine whether a user can access a record based on the value of a selected field.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Field-Based Access Control	Identify the target table and the field value that controls record access.
FR-2	Script-Controlled ACL	Create a scripted ACL that evaluates the field value against the user's access condition.
FR-3	User/Role Validation	Validate the logged-in user's role, identity, or other permitted criteria before granting access.
FR-4	Record Access Restriction	Allow access when the configured condition is satisfied and deny access when it is not satisfied.
FR-5	Unauthorized Access Handling	Prevent unauthorized users from viewing or accessing restricted records.
FR-6	ACL Testing and Validation	Test the ACL with authorized and unauthorized users and verify expected access behavior.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The ACL configuration and access rules should be clear, maintainable, and easy for administrators to understand.
NFR-2	Security	The solution must enforce record-level access restrictions and prevent unauthorized users from bypassing the ACL.
NFR-3	Reliability	The ACL should consistently return the correct access decision for valid and invalid field values.
NFR-4	Performance	ACL evaluation should execute efficiently without causing unnecessary delays when records are accessed.
NFR-5	Availability	Access control should remain available whenever users access the protected table and records.
NFR-6	Scalability	The design should support additional field-based conditions, roles, or access rules with minimal

		changes.